

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points. When writing proofs use COMPLETE SENTENCES! I will take away points otherwise.

Problem 1

Prove that the multiplicative identity is unique, that is $\exists! x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}$ then $x \cdot y = y$.

Problem 2

For each of the following two sets, find the maximum, minimum, infimum, and supremum if they are defined.

a) $\{\frac{1}{n} : n \in \mathbb{N}\}$

b) $\{x \in \mathbb{R} : x^2 < 2\}$

Problem 3

Prove that the natural numbers are closed under addition. That is, prove that if $m, n \in \mathbb{N}$, then $m + n \in \mathbb{N}$. (Hint: start by letting m be any natural, then induct on n .)

Problem 4

For each of the following statements, determine whether it is true or false and justify your answer.

- a) The set $\mathbb{Z}$ of integers is dense in $\mathbb{R}$.
- b) The set $\mathbb{R}^+$ of positive real numbers is dense in $\mathbb{R}$.
- c) The set $\mathbb{Q} - \mathbb{Z}$ of rational numbers that are not integers is dense in $\mathbb{R}$.

Problem 5

The Triangle Inequality states that $\forall a, b \in \mathbb{R}$ then $|a + b| \leq |a| + |b|$. Show that the Triangle Inequality becomes an equality if a and b are of the same sign.

Problem 6

Let $a > 0$. Prove that if x is a number such that $|x - a| < \frac{a}{2}$, then $x > \frac{a}{2}$.

Problem 7

Which of the following inequalities hold for all real numbers a and b . Justify your conclusion.

a) $|a + b| \geq |a| + |b|$.

b) $|a + b| \leq |a| - |b|$.

Problem 8

Prove that for all $n \in \mathbb{N}$,

$$\sum_{j=1}^n = \frac{n(n+1)(2n+1)}{6}.$$

Extra Credit:

Prove that given $x, y \in \mathbb{R}$, then $\forall n \in \mathbb{N}$

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$